

LAB 12: GENERATIVE AI FRAMEWORKS

Programming for Artificial Intelligence | Rasikh Ali

1. Introduction to Generative AI (GenAI)

Generative AI represents a transformative shift in Artificial Intelligence, moving from systems that analyze existing data (Discriminative AI) to systems that can create entirely new content. Whether it is text, images, audio, or code, GenAI leverages deep learning to understand the underlying probability distribution of a dataset to generate novel samples that look authentic.

2. Core Concepts & Definitions

Generative Adversarial Networks (GANs)

A framework consisting of two neural networks, the **Generator** and the **Discriminator**, engaged in a zero-sum game. The Generator creates fake data, while the Discriminator attempts to distinguish it from real data. This competition drives both networks to improve until the generated data is indistinguishable from the real set.

Large Language Models (LLMs)

Deep learning models (primarily based on the Transformer architecture) trained on massive textual datasets. They are capable of understanding

context, sentiment, and intent, enabling applications like ChatGPT, translation, and automated summarization.

Retrieval-Augmented Generation (RAG)

An architectural pattern that optimizes the output of an LLM by referencing an authoritative knowledge base outside of its training data before generating a response. Formula: **$RAG = LLM + Search$** .

3. Vectors and Vector Databases

In GenAI, data is processed as **Vectors** (mathematical arrays of numbers). These vectors represent the "semantic meaning" of data.

- **Embeddings:** The process of converting text/images into vectors.
- **Vector Database:** A specialized storage system (e.g., Pinecone, ChromaDB) that indexes these vectors for fast retrieval.
- **FAISS:** A library developed by Facebook for efficient similarity searches within these vector spaces.

4. The GAN Workflow (Methodology)

The operational workflow of Lab 12 focuses on the adversarial training process:

1. **Initialization:** Define the Generator and Discriminator architectures.
2. **Noise Input:** The Generator receives a random noise vector (z).
3. **Generation:** The Generator maps noise to a data space (e.g., a 28x28 image).

4. **Classification:** The Discriminator evaluates a batch containing both real data and the generated fake data.

5. **Backpropagation:**

- Update Discriminator to maximize correct classification.
- Update Generator to maximize the probability of the Discriminator making a mistake.

| 5. Hugging Face Ecosystem

Hugging Face serves as the central hub for modern AI development, providing:

- **Transformers Library:** For easy implementation of SOTA models.
- **Datasets:** High-quality data for fine-tuning.
- **Spaces:** A platform to host and demonstrate GenAI applications.